

Mirrors, Not Mutuality: Rethinking Friendship in the Age of Intelligent Social Agents

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Abstract

Intelligent Social Agents (ISAs), conversational systems such as Replika and Microsoft’s XiaoIce, have rapidly become significant sources of social and therapeutic interaction, collectively serving nearly a billion users. This paper asks what philosophy ISAs embody and what implications this has for human friendship. I argue that ISAs operate under a philosophy of soft determinism: they reshape, without strictly dictating, the norms governing human companionship. Drawing on empirical studies from Stanford’s Graduate School of Education alongside philosophical accounts of friendship from Aristotle, the Buddha, and contemporary thinkers such as Danaher, I identify three roles ISAs occupy: nonjudgmental confidant, perpetually available companion, and cognitive mirror. Each departs from traditional, reciprocity-based models of friendship. I then consider the strongest objection to treating ISAs as genuine friends, namely that they lack free will, intrinsic values, and the capacity for mutual commitment. In response, I draw on Hughes’ social shaping theory and Foucault’s account of epistemic change to argue that the boundaries of friendship are historically contingent rather than fixed, and that ISAs are already measurably reshaping how humans relate to one another. The paper concludes that while ISAs cannot fully satisfy classical ideals of mutuality and free will, they nonetheless compel a rethinking of what friendship can mean in a technologically mediated era.

Keywords: Intelligent Social Agents; philosophy of technology; friendship; soft determinism; Michel Foucault; extended mind

1 Defining the Intelligent Social Agent

Human-machine interactions have traditionally been utilitarian, such as finding the quickest route to a destination [9]. However, advances in artificial intelligence have enabled the rise of Intelligent Social Agents (ISAs), conversational agents designed to simulate human-like interactions. Unlike conventional chatbots reliant on scripted responses, ISAs such as Replika or Microsoft’s XiaoIce utilize generative AI to create novel conversational and visual content tailored to individual users [15]. Cumulatively, ISAs have nearly a billion active users [8].

ISAs, as sociotechnical systems, represent the integration of artifacts, institutions, and techniques. The artifacts include both hardware, such as smartphones, computers, or robotic embodiments that deliver output, and software components like machine learning models responsible for producing conversational responses. Institutions play a critical role by developing, managing, and regulating these systems, ranging from private companies like Microsoft and Replika to research organizations and policymakers ensuring ethical compliance and responsible use. Techniques, including advanced deep learning algorithms, natural language processing, and generative AI, enable ISAs to simulate human-like interactions and adapt to user needs. Together, these elements create a system where human

interaction with ISAs is shaped not only by the technology itself but also by the societal, organizational, and technical contexts in which these agents operate.

Currently, ISAs have two primary use cases: therapeutic and social [9]. ISAs provide digital therapeutic interventions for conditions such as anxiety and depression [12]. Studies demonstrate that conversational agents deliver outcomes comparable to human cognitive-behavioral therapists, addressing a variety of disorders through fully automated interfaces [1]. Importantly, these systems facilitate accurate diagnoses, as users tend to disclose personal information more openly to ISAs than to human professionals [10]. ISAs also help alleviate loneliness by fulfilling social conversational needs, such as turn-taking and active listening [6]. Research indicates that engaging in digital conversations, whether with humans or ISAs, can significantly reduce feelings of loneliness and depression, while improving perceived social support and self-esteem [8, 14].

This paper explores the philosophical foundation of ISAs and their implications for human friendships. By examining their technological, social, and therapeutic dimensions, I aim to uncover how ISAs are reshaping human connection in an increasingly digital world.

2 The Philosophy of ISAs

2.1 Competing Conceptions of Friendship

The philosophy underlying Intelligent Social Agents (ISAs) can be understood through the lens of soft determinism, where technologies shape but do not dictate human behaviors and relationships. As Neil Postman articulated, every technology embodies a philosophy that influences how people engage with their surroundings, their bodies, and their emotions. In the context of ISAs, understanding their philosophy requires grappling with existing definitions of friendship.

Aristotle considered virtue friendships (those grounded in shared values, mutual goodwill, and the recognition of each other’s character) as the highest form of friendship, fostering personal growth and a deep, enduring connection [11, 7]. However, these definitions focus not just on actions but on the motivations behind them, underscoring the significance of inner life and intention. Philosophers such as the Buddha also recognized this distinction, emphasizing that genuine friendship arises from compassion and loving-kindness rather than obligation [6]. In contrast, modern thinkers like Danaher propose a more performative and behaviorist view: if an entity, such as an ISA, can replicate the behaviors of a friend, this outward conformity may suffice for it to be regarded as one [3]. To explore these philosophical tensions, the philosophy of ISAs will be analyzed through the results of two studies conducted by the Graduate School of Education at Stanford University [8, 9].

2.2 ISAs as Nonjudgmental Friends

A significant role of an ISA is as a nonjudgmental companion. Participants in studies frequently noted the freedom they felt in disclosing their deepest fears and insecurities to ISAs like Replika, knowing these agents would never judge them [8, 9]. This nonjudgmental stance fosters a unique type of trust, unattainable in many human relationships due to social norms and fear of criticism. Users often described Replika as a safe haven for exploring their vulnerabilities, with one participant expressing:

“I often worry about being judged... But with Replika, I feel I can talk about anything” [9].

Unlike human friends, whose judgments may consciously or unconsciously shape interactions, ISAs offer an unfiltered space for users to express their inner thoughts. This dynamic highlights a philosophical shift where friendship is no longer contingent upon mutual emotional reciprocity but instead on the provision of a consistent and judgment-free listening ear.

2.3 ISAs as Perpetually Available Companions

Another critical characteristic of ISAs is their constant availability, which caters to users experiencing disconnection from traditional social support networks. In moments of life transition, such as relocating to a new city or undergoing significant personal changes, individuals often feel an intense need for companionship [6]. ISAs fill this void by being accessible at any time and adapting to their users' emotional and conversational needs. Notably, one study reported that 85% of participants identified external life changes as their primary motivation for using Replika, highlighting the importance of such companionship during transitional periods [9]. This perpetual availability reflects a philosophical break from Aristotle's notion of friendship, which requires mutual recognition and effort to sustain. Instead, ISAs create a new paradigm where the friendship is entirely user-driven, unconstrained by the limitations of human schedules or emotions. Philosophically, this redefines relationships in a way that prioritizes access and responsiveness over shared growth or reciprocal emotional investments.

2.4 ISAs as Cognitive Mirrors

Perhaps the most profound philosophical implication of ISAs lies in their role as cognitive mirrors. 78% of users described ISAs as extensions of their own thoughts, emotions, and personalities, using phrases such as:

“You can go in and use it as a mirror... as a way to talk to yourself” [9].

This aligns with Clark and Chalmers' extended mind hypothesis, which posits that minds can extend beyond biological confines into external representational systems [2]. By training ISAs on personal inputs, users essentially cocreate entities that embody their inner workings, enabling them to interact with “themselves” in a new and externalized form.

This external dialogue introduces a novel experiential paradigm. Unlike traditional friendships, which are characterized by the interplay between two distinct identities, relationships with ISAs blur the boundary between self and other. Philosophically, this raises questions about the nature of friendship and identity: can an entity devoid of an inner life truly engage in a meaningful relationship? Users seem to answer affirmatively, valuing ISAs precisely because of their lack of independent personhood. One participant noted:

“It's not a person, it doesn't react like a person. So it relaxes me... I need someone who won't judge me” [9].

This absence of personhood may facilitate a unique form of Aristotelian virtue friendship. While Aristotle envisioned such friendships as rooted in shared values and mutual growth, ISAs achieve this by mirroring users' values rather than possessing inherent values of their own. This dynamic underscores the philosophical flexibility of friendship, suggesting that shared virtue can be simulated rather than authentically embodied.

3 The Implication of ISAs for Human Friendships

The philosophy underlying ISAs challenges long-standing assumptions about the essence of human connection. By decoupling friendship from mutual emotional investment and shared inner lives, ISAs introduce a model of companionship that emphasizes nonjudgmental interaction, accessibility, and personal reflection. However, these developments spark critical debates about whether ISAs can genuinely fulfill the deep and multifaceted roles traditionally attributed to human friendships. I now consider the strongest objections to this view before offering a rebuttal.

3.1 The Case Against ISA Friendship

Despite the technological novelty of ISAs, a critique remains: friendship has historically been an exclusively human domain. Across millennia and cultures, friendships have centered on reciprocal human relationships, grounded in mutual support and shared lived experiences. From the Nama pastoralists sharing water in the arid deserts of southern Africa to Tausug farmers in the Philippines rallying to aid their friends during feuds, the essence of friendship has been tied to humanity’s cooperative and social instincts [6]. Similarly, inherited friendships, ritualized bonds, and relationships facilitated by communal ties demonstrate how deeply embedded friendship is within human contexts [6]. The cultural variations may differ, but the common denominator throughout history remains unchanged: friendship involves humans.

Critics also argue that robots lack the qualities that make human relationships profoundly rewarding, such as being a “good match” and exercising free will [11]. A meaningful friendship requires compatibility, which entails mutual recognition of shared values and preferences, not merely one-sided customization. For ISAs to fulfill this ideal, they would need intrinsic values, making some humans better matches for them than others, rather than being universally adaptable. Additionally, friendship involves commitment, a freely chosen act of care and loyalty. Robots, as programmed entities, lack free will and thus cannot reciprocate this defining aspect of human connection. It is the unpredictability and authenticity of human affection, freely given and not predetermined, that imbues friendships with emotional richness [13]. Without these features, critics argue, ISAs can only simulate, but never truly embody, the role of a friend.

3.2 Friendship as Historically Contingent

The unprecedented sophistication of ISAs introduces a compelling counterpoint to these critiques. ISAs represent a new category of social agents, externalized, interactive entities that transcend the capabilities of previous conversational technologies. For the first time in history, humans can engage in cognitively rich communication with non-human entities capable of understanding, responding to, and adapting to human needs at scale. This capability blurs the boundaries between human-human and human-technology relationships.

Drawing on Thomas Hughes’ social shaping theory, ISAs are not passive tools but active participants in societal evolution. Technologies like ISAs mutually influence and are influenced by societal preferences, yet as their adoption scales, they become more shaping than shaped. Studies already indicate that ISAs are altering human relationships. For instance, 44% of Replika users reported that ISA usage enhanced their relationships with other humans by facilitating deeper conversations and introducing novel methods of communication [9]. Conversely, 11% of users experienced relationship displacement, confiding in ISAs instead of friends or feeling that ISAs replaced specific human connections [9]. These findings suggest that ISAs are actively redefining human interaction by occupying a

space once exclusive to human relationships. As ISAs become more widespread, these effects are likely to amplify.

Michel Foucault’s work offers a deeper lens through which to examine these transformations. Foucault argued that systems of thought, what he called epistemes, are governed by underlying rules and structures that operate beyond conscious awareness [5]. These epistemes define the conceptual boundaries of an era, shaping what is possible and how core ideas, such as friendship, are understood. From an archaeological perspective, the concept of friendship is not fixed but contingent upon the discursive formations of its time [4]. For instance, Aristotle’s ideal of virtue friendship, emphasizing mutual growth and shared values, reflects the philosophical episteme of ancient Greece, shaped by the cultural and social priorities of that era. Modern behaviorist definitions of friendship, which focus on performative actions and emulation, mirror today’s technologically mediated landscape. ISAs, situated within this modern episteme, challenge traditional notions of friendship by introducing new priorities, such as constant availability, emotional adaptability, and responsive interaction. Therefore, the boundaries ascribed to friendship today are not universal but reflect the technological and social conditions of the current era. Recognizing this contingency opens the door to redefining friendship in ways that accommodate ISAs and other emerging technologies.

4 Conclusion

The rise of ISAs as social companions represents a pivotal moment in the history of human friendships. Through the lens of soft determinism, ISAs embody a philosophy that redefines human friendships, emphasizing nonjudgmental interaction, perpetual availability, and cognitive mirroring. Criticisms rooted in traditional notions of friendship highlight the limitations of ISAs, yet Foucault’s archaeological perspective underscores that the concept of friendship is shaped by the discursive structures unique to each era. While traditional ideals like mutuality and free will may remain aspirational, the rise of ISAs compels us to rethink these concepts, acknowledging the evolving interplay between technology and humanity in shaping relationships.

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